# Similarity, Ratio, Area & Centroid Structure

Integrated Grade-9 learner pack

# Similarity, Ratio, Area & Centroid Structure

## The governing question

Most problems here become short after one choice: what relation is actually preserved? Do not start by calculating every length. Prove the structure first, then transfer only the information the structure licenses.

## 1. Similarity is a proof, not a visual impression

Two triangles are similar when corresponding angles match and corresponding sides have one common scale factor. Standard routes are AA, SAS and SSS. The correspondence order matters: if A <-> P, B <-> Q, C <-> R, then write ratios in that order.

A common trap is to upgrade similarity to congruence. Equal angles determine shape, not size. Congruence is the special case where the scale factor is 1.

### A quick contrast

Triangles with sides 3,4,5 and 6,8,10 are similar but not congruent. Triangles with sides 3,4,5 and 3,4,5 are both similar and congruent.

## 2. Decide whether the target is linear or area

If similar figures have length scale factor k, then corresponding lengths and perimeters scale by k, while areas scale by k^2.

This is not a memorization trick. If both base and height are multiplied by k, the factor 1/2 * base * height picks up two factors of k.

### Reverse direction

If two figures are already known to be similar and their areas are in ratio 25:49, the positive side ratio is 5:7. Without known similarity, equal or proportional areas alone do not determine the side ratio.

## 3. Area ratios can work without similarity

For triangles sharing the same altitude, area ratio equals base ratio. For triangles sharing the same base, area ratio equals height ratio. These are often cheaper than proving similarity.

Suppose points P and Q lie on the same line BC. Triangles ABP and ABQ have the same altitude from A to line BC, so [ABP]:[ABQ]=BP:BQ.

## 4. Parallel lines are ratio-transfer machines

If D lies on AB, E lies on AC, and DE is parallel to BC, then triangle ADE is similar to triangle ABC. One proved parallel relation transfers the division ratio across the two sides:

AD/AB = AE/AC = DE/BC.

If AD/AB=2/5, the area ratio is not 2/5; it is 4/25.

## 5. The centroid: use the invariant that matches the target

The three medians meet at the centroid G. Along a median from vertex A to midpoint M of BC,

AG:GM = 2:1.

For area questions, an even more useful statement is

[GAB] = [GBC] = [GCA] = (1/3)[ABC].

All three medians together split the whole triangle into six equal-area small triangles. These facts are shape-independent; no side lengths or angles are required.

Coordinates can verify the centroid, but do not reach for a coordinate formula when the area share closes the problem in one line.

## 6. Area decomposition: compute fractions before lengths

When lines cut a figure into pieces, label each piece as a fraction of a convenient whole. Add or subtract only after the fractions are controlled by similarity, common altitude/base, midpoint, or centroid facts.

This avoids a frequent dead end: computing several side lengths merely to feed an area formula when the entire answer is already determined by ratios.

## 7. Two validated historical mechanisms

- IOQM-2024-Q12: a square with side-trisection points and two crossing lines. A ratio/coordinate intersection gives the target triangle area. The independently recomputed answer is 96.

- IOQM-2023-Q05: medians, their centroid and midpoints on the medians. The requested area is a shape-independent fraction of the whole; the independently recomputed answer is 10.

Both source problems are text-only in the frozen corpus; no historical figure is reproduced here.

## 8. Representation choice

Use a synthetic ratio route when parallels, midpoint relations or obvious similar triangles already control the picture. Use coordinates when several line intersections are numerical and a clean placement turns them into two linear equations. A coordinate solution is not automatically more rigorous or more advanced; it is simply another representation. The best route is the one with the shortest justified chain.

## 9. A complete example

In triangle ABC, point D lies on AB with AD:DB=3:2, point E lies on AC, and DE is parallel to BC. If [ADE]=54, find [ABC].

AD/AB = 3/5, so the area scale is $(3/5)^2=9/25$. Therefore [ABC]=54*(25/9)=150.

Notice the sequence: prove/recognize similarity -> identify linear scale -> square because the target is area -> solve.

## 10. Self-check questions before you calculate

- What exactly proves the triangles similar?

- Have I written the correct vertex correspondence?

- Is the target a length/perimeter or an area?

- Do the compared triangles share a base or altitude?

- Is a median/centroid fact enough without coordinates?

- Is a line really parallel, or does it merely look parallel?

- Can I express the desired region as a fraction of a larger area first?

# First-Step Reference

When the picture looks busy, ask these in order.

| Situation | First step | Do not do first |
|---|---|---|
| two angle pairs match | state AA similarity and correspondence | claim congruence |
| side triples are proportional | state SSS similarity and the scale | compare random noncorresponding sides |
| two proportional side pairs plus equal included angle | state SAS similarity | use SAS when the equal angle is not included |
| a segment is parallel to one side of a triangle | identify the small/large similar triangles | copy a ratio just because lines look parallel |
| similar figures, length target | use scale k | square the ratio |
| similar figures, area target | use scale k^2 | use the side ratio unchanged |
| same altitude | compare bases | prove similarity unnecessarily |
| same base | compare heights | compute both full areas separately |
| centroid/medians, area target | use one-third or six-equal-area structure | start with coordinate formulas |
| ratio-only line intersection | build a ratio/area decomposition | compute every length |
| intersection arithmetic is cleaner in a square/rectangle | place simple coordinates | force a long synthetic chase |

## One-sentence router

Prove the relation, decide linear-versus-area, exploit shared height/base or centroid structure, then change representation only if it shortens the justified work.

# Recognition and First-Line Lab

Answer each before reading any explanation. State the first justified relation, not a guessed formula.

**1. Two triangles have corresponding angles 42 degrees, 68 degrees and 70 degrees. What can you conclude if no side length is given?**

**2. One triangle has sides 6, 8, 10 and another has sides 9, 12, 15. Classify the relation.**

**3. Similar triangles have corresponding side ratio 4:7. What is their area ratio?**

**4. Two triangles share the same altitude and have bases 5 and 8. What is their area ratio?**

**5. G is the centroid and M is the midpoint of BC. What is AG:GM?**

**6. A median divides a triangle into two parts. How do their areas compare?**

**7. In triangle ABC, DE is parallel to BC with D on AB and AD/AB=2/5. What is DE/BC?**

**8. Two similar figures have linear scale 2:3. What is the area scale?**

**9. Two pairs of sides are proportional, but no included-angle equality and no third-side ratio is known. Is similarity established?**

**10. Two triangles have three corresponding sides equal. Are they similar, congruent, or both?**

**11. G is the centroid of triangle ABC. What fraction of the whole area is triangle GBC?**

**12. Two triangles have equal area. Must they be similar?**

## First-line discipline

For every item above, a valid first line names the licensing structure: AA/SAS/SSS similarity, a shared altitude/base, a proved parallel relation, or centroid structure. A diagram alone is never the license.

# Practice and Transfer Bank

Attempt each problem before consulting the teacher key. The section names describe the learning progression; no method is supplied inside an item.

## Reconnect

**1. In triangle ABC, D lies on AB and E on AC with DE parallel to BC. If AD=6, DB=3 and AC=15, find AE.**

**2. Triangles are similar. A side 7 in the smaller corresponds to 14 in the larger. What length corresponds to side 9 in the smaller?**

**3. The side scale from one similar triangle to another is 3:4. Give the area ratio in the same order.**

## Structure with support

**4. Two similar triangles have areas 81 and 144. A corresponding side in the smaller is 12. Find the corresponding side in the larger.**

**5. Two triangles have the same altitude. Their bases are 13 and 7. If the first area is 52, find the second area.**

**6. Two triangles have the same base. Their perpendicular heights are 9 and 4. If the first area is 63, find the second area.**

**7. M is the midpoint of BC in triangle ABC. If the area of ABC is 126, find the area of ABM.**

## Reduced support

**8. G is the centroid of triangle ABC. If the area of ABC is 150, find the area of GBC.**

**9. The three medians of a triangle form six equal-area small triangles. If each has area 14, find the area of the original triangle.**

**10. In triangle ABC, DE is parallel to BC and AD/AB=2/5. Find area(ADE):area(ABC).**

**11. In the setting above, if the area of ABC is 200, find the area of ADE.**

## Mixed independent

**12. A square ABCD has side 12. Points E,F trisect CD in the order C-E-F-D. Lines BF and AE meet at M. Find the area of triangle MAB.**

**13. In triangle ABC, E and F are the midpoints of AC and AB; medians BE and CF meet at G. Y and Z are the midpoints of BE and CF. If area(ABC)=240, find area(GYZ).**

**14. Point P lies on BC with BP:PC=2:5. If area(ABP)=18, find area(APC).**

**15. G is the centroid on median AM. If AG=10, find AM.**

## Transfer

**16. In triangle ABC, D lies on AB and E on AC, DE parallel to BC, and AD:DB=3:2. If area(ADE)=54, find area(ABC).**

**17. In triangle ABC, D and E divide AB and AC with AD:DB=AE:EC=1:2, and DE is parallel to BC. If area(ABC)=90, find the area of quadrilateral DBCE.**

**18. Two similar triangles have scale factor 5/3 from smaller to larger. If the smaller perimeter is 36, find the larger perimeter.**

# Mixed Mastery Test

First attempt: no hints, no method labels, no worked examples. Show enough reasoning to identify the structure you used.

**1. In triangle ABC, D lies on AB and E lies on AC with DE parallel to BC. If AD=8, AB=20 and AE=12, find AC.**

**2. Two similar triangles have areas in the ratio 25:49. A side of the larger triangle is 21. Find the corresponding side of the smaller triangle.**

**3. Two triangles have all three corresponding angles equal, but one is visibly larger. Classify the strongest guaranteed relation.**

**4. Two triangles share an altitude. Their bases are in the ratio 4:7. If the smaller area is 32, find the larger area.**

**5. G is the centroid on median AM and AM=18. Find AG and GM.**

**6. G is the centroid of triangle ABC with area 324. Find the area of triangle ABG.**

**7. D is the midpoint of BC and E is the midpoint of AC. If area(ABC)=200, find area(CDE).**

**8. A square ABCD has side 20. Points E,F trisect CD in the order C-E-F-D. Lines BF and AE meet at M. Find area(MAB).**

**9. In triangle ABC, E and F are midpoints of AC and AB; medians BE and CF meet at G. Y and Z are midpoints of BE and CF. If area(ABC)=960, find area(GYZ).**

**10. Point D lies on BC. If area(ABD):area(ADC)=7:4, find BD:DC.**

**11. The length scale from a smaller similar triangle to a larger one is 5/2. If the smaller area is 32, find the larger area.**

**12. Two triangles have two pairs of proportional sides, but the included angles are 50 degrees and 60 degrees. Can similarity be concluded?**